Softmax: loss precision and epsilon matrix

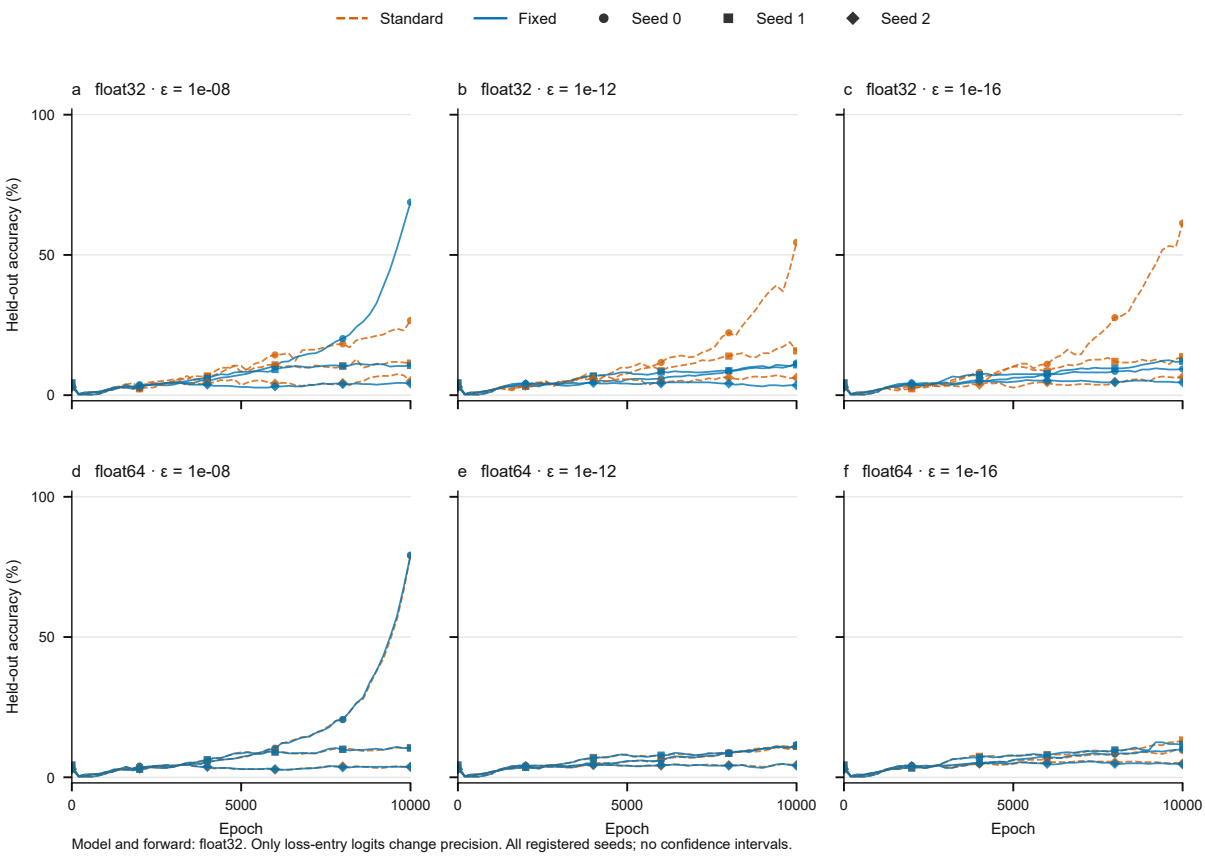


Figure 1: All seeds and both repair arms across the supported matrix.

Loss precision × epsilon: paired reconstruction

All 36 specified training trajectories completed, with independently checked final models and gradient endpoints.

Observed result: fixed CE is higher in 5/18 paired endpoints, lower in 12/18, and tied in 1/18. The best final held-out accuracy across all arms is 79.210%. These runs do not reproduce the paper's high-generalization table and do not establish a uniform finite-budget generalization benefit.

At the model interface, the fixed arm's maximum target-zero count is 0 across 918 producer-recorded checkpoints; standard CE reaches 243/480. Final gradient counts are independently reloaded; intermediate counts are recorded by the producer.

Locally registered follow-up; three seeds, float32 model/forward, two supported loss precisions and three epsilon values. Not an exact paper reproduction. bfloat16 remains UNQUALIFIED.

All cells below are retained. Delta is fixed minus standard in percentage points; no population inference or significance test.

Loss dtype	Epsilon	Seed	Standard test (%)	Fixed test (%)	Delta (pp)
float32	1e-08	0	26.611	68.815	+42.204
float32	1e-08	1	11.435	10.603	-0.832
float32	1e-08	2	4.990	4.158	-0.832
float32	1e-12	0	54.470	11.435	-43.035
float32	1e-12	1	15.800	10.811	-4.990
float32	1e-12	2	6.237	3.534	-2.703
float32	1e-16	0	61.331	9.356	-51.975
float32	1e-16	1	13.721	12.058	-1.663
float32	1e-16	2	6.445	4.574	-1.871
float64	1e-08	0	79.002	79.210	+0.208
float64	1e-08	1	10.603	10.395	-0.208
float64	1e-08	2	3.742	3.742	+0.000
float64	1e-12	0	11.019	11.642	+0.624
float64	1e-12	1	11.019	11.227	+0.208
float64	1e-12	2	4.366	4.158	-0.208
float64	1e-16	0	9.563	9.979	+0.416
float64	1e-16	1	13.306	11.642	-1.663
float64	1e-16	2	4.990	4.574	-0.416

Evidence and limitations

Six float32/epsilon=1e-8 final model hashes exactly replay the parent experiment. All cells reuse that parent's original initial models and splits.

Each trajectory records 51 checkpoints, including actual training-batch target-gradient zero counts before and after the loss-to-model precision boundary. Final counts are independently replayed from all 36 saved models on the recorded CUDA runtime. Intermediate counters remain producer logs, not an independently reloaded trajectory.

Final held-out predictions, margins and float64 cross-entropy are independently recomputed from raw saved tensors by NumPy. Cross-entropy comparisons use relative error with no absolute floor.

The full loss-precision/epsilon grid of supported kernels is covered; the paper's bf16 grid, additional baselines, unspecified architecture/initialization choices and further tasks remain outside this reconstruction. Generalization can depend strongly on seed and implementation choices.

Source paper, section 6 and Table 1. The protocol and source hashes define this local study; completed cells are not selected for favorable outcomes.

Downloads: trajectories, paired endpoints, protocol, raw evidence, final gradient audit, source and checkpoints.

Publication masters: editable SVG, PDF at 183 mm.

CPU acceptance: `python reference/softmax_matrix/verify.py`. Retraining needs the recorded CUDA runtime and a separately registered output location; the original producer refuses overwrite.